

SEAGAS

Daily Development Report

28 September 2026 · Sprint 2

Unit	COS40005 Computing Technology Project A
Group	Group 10
Repo	https://github.com/j23039278-netizen/Group10_ProjectA
Branch	dev
Date	28 September 2026

This report covers: (1) What was completed today, (2) Test results, (3) Files changed, (4) Step-by-step guide for all teammates to pull and run the latest code.

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1. Today's Progress Overview

Task	Who	Result
Expand skills_library 32 → 101 skills	Carl	■ Success
Fix jobs.py skill sort order bug	Carl	■ Fixed
Merge fix/requirements PR (Yong Hin)	Carl	■ Merged
Run seed_job_role_skills.sql (79 mappings)	Carl	■ Success
Build React frontend — 7 pages	Carl	■ Success
Connect frontend to backend API (CORS fix)	Carl	■ Working
End-to-end test: Login → Assessment → Skill Gap	Carl	■ Verified

All core features are now working end-to-end. Students can login, run assessments, and view skill gaps.

2. What Was Done & How

2.1 Skills Library: 32 → 101 Skills

New file: database/seed_skills_library_v2.sql — added 69 new skills across all 4 categories.

Category	Notable New Skills Added	Total
Technical	Wireshark, Nmap, Kali Linux, SIEM, Metasploit, Burp Suite, Kubernetes, Terraform, Pandas, NumPy, Matplotlib	53
AI_Digital	GCP, MLOps, Generative AI, TensorFlow, PyTorch, Cybersecurity, Network Security, Ethical Hacking, Incident Response	25
Analytical	Critical Thinking, Business Analysis, Risk Analysis, Data Modelling, A/B Testing, Financial Analysis	12
Soft	Adaptability, Project Management, Attention to Detail, Self-Motivation, Documentation	11
TOTAL	101 skills now in database	101

2.2 Bug Fixes

Bug 1 — Skill sort order (jobs.py): 'bonus' was sorting before 'required' alphabetically. Fixed with CASE WHEN.

```
# File: backend/routers/jobs.py

# Before (WRONG – bonus sorts before required alphabetically):
```

```
ORDER BY jrs.importance, sl.category, sl.skill_name

# After (CORRECT — explicit priority ordering):

ORDER BY

CASE jrs.importance

WHEN 'required' THEN 1

WHEN 'preferred' THEN 2

WHEN 'bonus' THEN 3

END,

sl.category, sl.skill_name
```

Bug 2 — CORS blocking React (main.py): Browser blocked frontend (port 5173) from calling backend (port 8000).

```
# File: backend/main.py

# Before:

allow_origins=["http://localhost:3000"],

# After:

allow_origins=["http://localhost:3000", "http://localhost:5173"],
```

2.3 React Frontend Built (7 Pages)

Page / File	What it does
Login.jsx	Login with 3 roles (Student / Advisor / Admin), JWT token stored
Dashboard.jsx	Job Readiness Score, 7-dimension breakdown, quick actions, assessment history
Profile.jsx	4 tabs — Academic info, Skills (from library dropdown), Projects, Certifications
JobAnalysis.jsx	Select role template OR paste custom JD → Run Assessment → see results instantly
SkillGap.jsx	Full skill breakdown: 7-dimension scores + Strong / Developing / Gap columns
Recommendations.jsx	Shows courses/certs/projects per skill gap (needs DB data from Aaron)
AdvisorDashboard.jsx	3 tabs: Cohort overview + bar chart, Skill gap table, Full student list

3. Test Results

Test account: student0001@synthetic.example.com / Seagas@2026

Step	Action	Result
1	Open http://localhost:5173	■ Login page appears
2	Login as student0001	■ Dashboard loads — Aryan Maharaj, Data Scientist, GPA 3.63
3	Job Analysis → Cybersecurity Analyst → Run Assessment	■ Score: 58.33%
4	View Skill Gap page	■ 7-dimension scores + 3 columns displayed
5	Strong Skills	■ Azure, Cloud Computing, Git, Problem Solving, Python, SQL
6	Skill Gaps	■ AWS, C++, Communication, Linux
7	Recommendations page	■■ Empty — recommendations_library needs data (Aaron's task)
8	Advisor login	■■ Pending — advisor account not created yet

4. Files Changed Today

File	Change Type	What Changed
database/seed_skills_library_v2.sql	NEW	69 new skills added
backend/routers/jobs.py	FIXED	CASE WHEN sort order for importance
backend/main.py	FIXED	CORS added port 5173
frontend/src/api/client.js	NEW	All API calls (auth, students, jobs, assessments, advisor)
frontend/src/App.jsx	NEW	Router + role-based routing
frontend/src/App.css	NEW	Global styles
frontend/src/pages/Login.jsx	NEW	Login page
frontend/src/pages/Dashboard.jsx	NEW	Student dashboard
frontend/src/pages/Profile.jsx	NEW	Profile management (4 tabs)
frontend/src/pages/JobAnalysis.jsx	NEW	Job analysis + assessment runner
frontend/src/pages/SkillGap.jsx	NEW	Skill gap analysis page

frontend/src/pages/Recommendations.jsx	NEW	Recommendations page
frontend/src/pages/AdvisorDashboard.jsx	NEW	Advisor cohort dashboard
frontend/src/components/Sidebar.jsx	NEW	Navigation sidebar

5. Teammate Setup Guide

IMPORTANT: Pull repo alone is NOT enough. You must also run 2 SQL files and install frontend dependencies. Follow ALL steps below.

1 Pull Latest Code from GitHub

```
git checkout dev  
  
git pull origin dev
```

Make sure you see the latest files in database/ and frontend/src/

2 Run New SQL Seed Files (Database)

You need to run these 2 files to update your local database:

```
# From project root folder (Group10_ProjectA):  
  
psql -U postgres -d seagas_db -f database/seed_skills_library_v2.sql  
  
# Expected: INSERT 0 69  
  
psql -U postgres -d seagas_db -f database/seed_job_role_skills.sql  
  
# Expected: INSERT 0 79
```

If you see 'INSERT 0 0' instead of 'INSERT 0 69' or 'INSERT 0 79', the data already exists — that is fine, ON CONFLICT DO NOTHING handles it.

3 Install Frontend Dependencies

node_modules is not in GitHub — you must install it locally:

```
cd frontend  
  
npm install  
  
# Wait for installation to complete (~30 seconds)
```

4 Create Your .env File (if not already done)

The .env file is NOT in GitHub. Create it manually in the backend/ folder:

```
# File: backend/.env  
  
DATABASE_URL=postgresql://postgres:YOUR_PASSWORD@localhost:5432/seagas_db  
  
SECRET_KEY=seagas_super_secret_key_change_in_production_2026
```

```
ALGORITHM=HS256
```

```
ACCESS_TOKEN_EXPIRE_MINUTES=30
```

Replace YOUR_PASSWORD with your PostgreSQL password set during installation.

5 Start Backend (Terminal 1)

```
cd backend

uvicorn main:app --reload

# OR use Yong Hin's one-click script:

# Double-click start_backend.bat in the backend/ folder

# Expected output:

# INFO: Uvicorn running on http://127.0.0.1:8000

# INFO: Application startup complete.
```

6 Start Frontend (Terminal 2 — New Terminal)

```
cd frontend

npm run dev

# Expected output:

# VITE v5.x ready in xxx ms

# Local: http://localhost:5173/
```

7 Open Browser & Test Login

Open: <http://localhost:5173>

```
Test Account:

Email: student0001@synthetic.example.com

Password: Seagas@2026
```

You should see the Dashboard with GPA and Target Role after login.

Quick Check: If login fails → make sure backend is running at port 8000. If page is blank → check frontend terminal for errors. If 'No required skills' → you forgot Step 2.

6. What's Next — Sprint 2 Remaining Tasks

6.1 High Priority

Task	Who	Why Important	How
Populate recommendations_library	Aaron	Recommendations page is empty	Run INSERT SQL for each skill → course/cert/project
NLP module (spaCy + Sentence Transformers)	Yong Hin	Core AI feature, needed for evaluation	Create nlp/extractor.py and nlp/matcher.py
Create advisor account	Anyone	Can't test Advisor Dashboard	POST /api/auth/register with role='advisor'
Test Profile page	All	Skills need to be added before assessment	Login → My Profile → Skills tab → Add skills

6.2 How to Create Advisor Account

Go to <http://127.0.0.1:8000/docs> → POST /api/auth/register → Try it out → Enter:

```
{
  "email": "advisor@seagas.com",
  "password": "Advisor@2026",
  "full_name": "Dr. Rashid",
  "role": "advisor"
}

# Then login at http://localhost:5173
# Select "Advisor" role button before signing in
```

6.3 How to Populate Recommendations (Aaron)

Run SQL like this for each skill gap:

```
-- Example: Add resource for Power BI skill gap

INSERT INTO recommendations_library
(skill_id, resource_title, resource_type, provider, url, duration_hours, cost, difficulty)
SELECT sl.skill_id,
'Microsoft Power BI Fundamentals',
'course', 'Microsoft Learn',
```

```
'https://learn.microsoft.com/en-us/training/powerbi/',
8, 'Free', 'beginner'

FROM skills_library sl WHERE sl.skill_name = 'Power BI';

-- Repeat for: AWS, Linux, C++, Communication, Docker, Nmap, Wireshark etc.

-- Aim for at least 2 resources per skill
```

6.4 Sprint 2 Status

Task	Who	Priority	Status
Recommendations library populated	Aaron	■ High	Not started
NLP module (keyword + semantic)	Yong Hin	■ High	Not started
Create advisor account + test dashboard	Anyone	■ Medium	Pending
All 5 members pull + test frontend	All	■ Medium	Pending
Expand skills_library to 150	Yong Hin	■ Medium	101 done, 49 more
Student development tracking UI	Aaron	■ Low	Not started
System integration testing	All	■ Sprint 3	Week 11-12
AI evaluation (L1 vs L2 comparison)	Yong Hin	■ Sprint 4	Week 13-14